

Geometry
Unit 3
Lesson 1 Homework

Name **Answer Key** _____
Date _____
Period _____

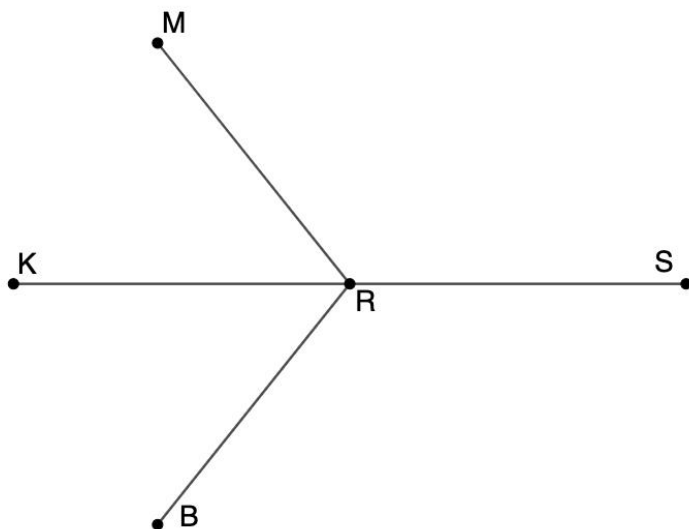
Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank

Definition of Linear Pair	Linear Pair Postulate
Supplements of congruent angles are congruent.	Complements of congruent angles are congruent.
Angle addition postulate	Substitution property of equality
Definition of supplementary angles	Definition of complementary angles

Given: $\angle KRB \cong \angle KRM$

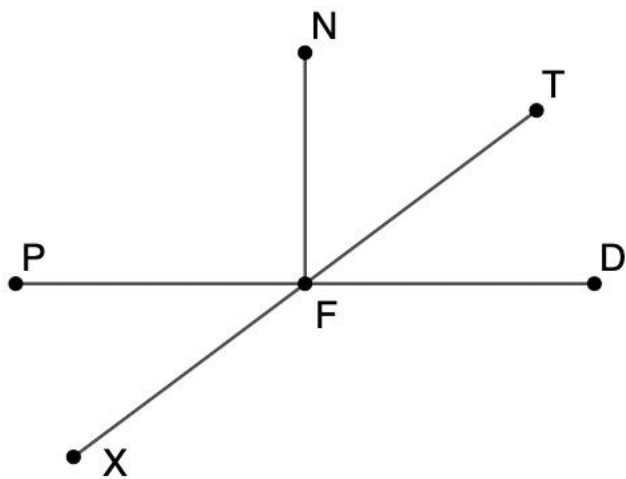
Prove: $\angle BRS \cong \angle MRS$



Complete the table using the word bank to prove $\angle BRS \cong \angle MRS$.

Statement	Reason
$\angle KRB \cong \angle KRM$	Given
$\angle KRB$ and $\angle SRB$ form a linear pair	1. Definition of Linear Pair
$\angle KRM$ and $\angle MRS$ form a linear pair	2. Definition of Linear Pair
$\angle KRB$ and $\angle SRB$ are supplementary angles	3. Linear Pair Postulate
$\angle KRM$ and $\angle MRS$ are supplementary angles	4. Linear Pair Postulate
$\angle BRS \cong \angle MRS$	5. Supplements of congruent angles are congruent.

Line segments $\overline{PD}$, $\overline{NF}$, and $\overline{TX}$ intersect at point F as shown where $m\angle PFN = 90^\circ$, $m\angle XFP = (3x - 56)^\circ$, $m\angle NFT = (4z + 7)^\circ$ and $m\angle TFD = (x + 2)^\circ$.



6. Find the value of x .

$$x = 29$$

7. Find the value of z .

$$z = 13$$

8. What is $m\angle NFT$?

$$m\angle NFT = 59^\circ$$

9. What is $m\angle XFD$?

$$m\angle XFD = 149^\circ$$